IITM PRAVARTAK
CATALYSING INNOVATION

Professional Certificate Programme in Agentic AI and Applications

Agentic Tools in Python

Revisit the Learning So Far

Let’s take a quick look at the key ideas and concepts you have explored in your learning journey.

- How embeddings capture meaning by converting text into numerical vectors
- Distinction between token vectors and sentence embeddings and when each is applied
- How cosine and Euclidean distance measure similarity between vectors
- How nearest neighbours are identified in vector space to find related text
- How embeddings can be visualised using dimensionality reduction techniques
- How vector search uses embeddings to retrieve relevant information
- A journaling activity to reflect on key insights about embeddings

Topics Covered

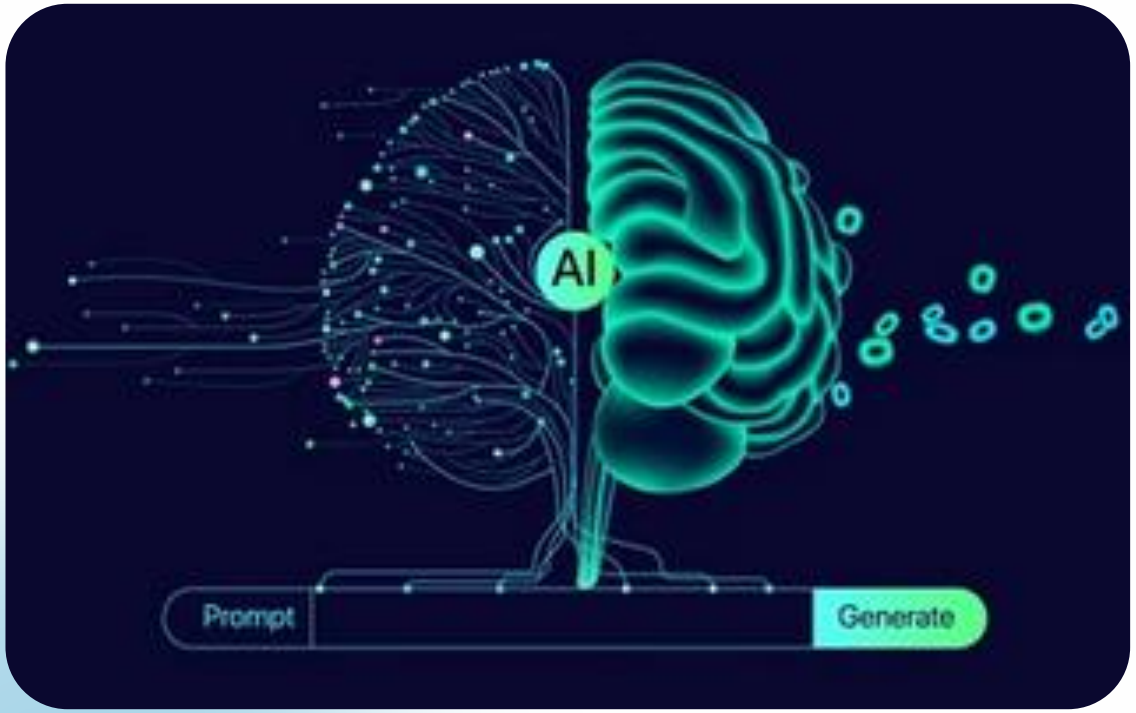
- Introduction to LangChain
- Prompt Formatting & Response Parsing
- Prompt Formatting
- Response Parsing
- LangChian Tools & Tool Calling
- Introduction to Prompt Templates

Introduction to LangChain

Building applications with Large Language Models (LLMs) that are practical, powerful and production-ready for real-world deployment.

Why LangChain

LLMs are Powerful but Limited



Context window constraints
Struggle with long contexts and lose information beyond token limits

No memory persistence
Cannot retain details from past interactions or conversations.

Data access limitations
No direct access to external databases, APIs or real-time data

Why LangChain?

LangChain solves this by:

- Adding persistent memory
- Connecting to data sources
- Orchestrating multiple components

Core Idea of LangChain

LangChain is a framework that connects isolated LLMs into intelligent systems for real-world use.

Connect LLMs with external tools	Manage chains of reasoning	Store and retrieve knowledge	Enable real-world AI applications
Integrate APIs, databases, web search and custom functions.	Handle sequential processes and complex workflows	Support persistent memory and document search	Build production-ready, scalable AI systems.

Main Components of LangChain

LLMs

Core language models from OpenAI, Hugging Face, Anthropic and others

Prompts

Structured templates to optimise and standardise model inputs

Chains

Sequential workflows combining models, tools and processes

Agents

Autonomous systems that select actions and tools based on context

Memory

Persistent storage of conversation history and context

VectorStores

Efficient systems for semantic search and document retrieval

How it Works

Basic Flow

User input
Natural language query or command

Prompt template
Structure and format the input

LLM processing
Generate intelligent response

Output
Final result to user

How it Works

Enhanced with LangChain

Memory integration
Conversation-aware applications that remember context

Document retrieval
Search company documents, PDFs and databases

Example Use Cases

Intelligent chatbots

Provide context-aware customer service, remembering conversation history and user preferences.

Document Q&A systems

Answer questions using company documents, legal contracts and knowledge bases.

Code assistants

Support development by connecting to APIs, documentation and version control

Example Use Cases

Research assistants

Retrieve and synthesise information from scientific papers

RAG applications

Combine real-time data retrieval with AI-generated content

Simple Demo (Conceptual)

```python
from langchain import OpenAI, ConversationChain

# Initialize the LLM
llm = OpenAI(temperature=0)

# Create a conversation chain
conversation = ConversationChain(llm=llm)

# Interactions
response1 = conversation.run("Hello, who won the 2018 FIFA World Cup?")
print(response1)  # Expected: "France won the 2018 FIFA World Cup"

response2 = conversation.run("What was the final score?")
print(response2)  # LangChain remembers context and responds appropriately
```

Simple Demo (Conceptual)

Key Difference

Without Langchain	With LangChain
LLM answers once, no context retention	App maintains conversation memory and context awareness

Architecture Overview

Input layer
Receive user prompts, queries and natural language commands

Core processing
Orchestrates workflows with Chains and enable dynamic decisions via Agents

Memory layer
Stores conversation history and contextual state persistently

Data layer
Uses vector databases like Pinecone, FAISS and Chroma for semantic search

Output layer
Delivers structured responses, actions and final results to users.

Ecosystem & Integrations

Vector databases
FAISS, Pinecone, Weaviate, Chroma for similarity search

LLM providers
OpenAI, Hugging Face, Anthropic, Cohere, local models

Monitoring
LangSmith for debugging and performance optimisation

External tools
Google Search, APIs, SQLdatabases and web scraping tools

Benefits

Rapid prototyping
Builds AI applications swiftly using pre-built components and detailed documentation

Modular Flexibility
Plug-and-play architecture allows easy swapping of models, tools, and data sources

Scalable architecture
Supports both small experiments and large-scale production deployments seamlessly

Rich ecosystem
Offers extensive integrations with popular tools, databases and AI services

Limitations & Challenges

Adds unnecessary complexity for simple use cases without advanced needs

Depends on external vector databases, APIs and third-party services

Requires frequent updates due to fast-changing library and maintenance

Complexity overhead

External dependencies

Rapid evolution

Consider the trade-offs: Assess if LangChain's full framework is necessary or if simpler solutions suffice.

Demo: Set up LangChain and OpenAI integration, execute a simple LLMChain

Prompt Formatting & Response Parsing

Designing inputs and structuring outputs for reliable LLM use

Building Reliable AI Systems

Challenge
LLMs produce varied natural language responses, complicating workflows needing consistent, structured outputs

Solution
Apply strategic prompt formatting and robust parsing to ensure reliable, deployable AI applications

Why Structure Matters

Natural language output
LLMs generate varied, human-like responses that differ in format, length and structure each time.

Production requirements
Applications need predictable, parseable data formats for APIs, databases and user interfaces.

Why Structure Matters

Prompt formatting

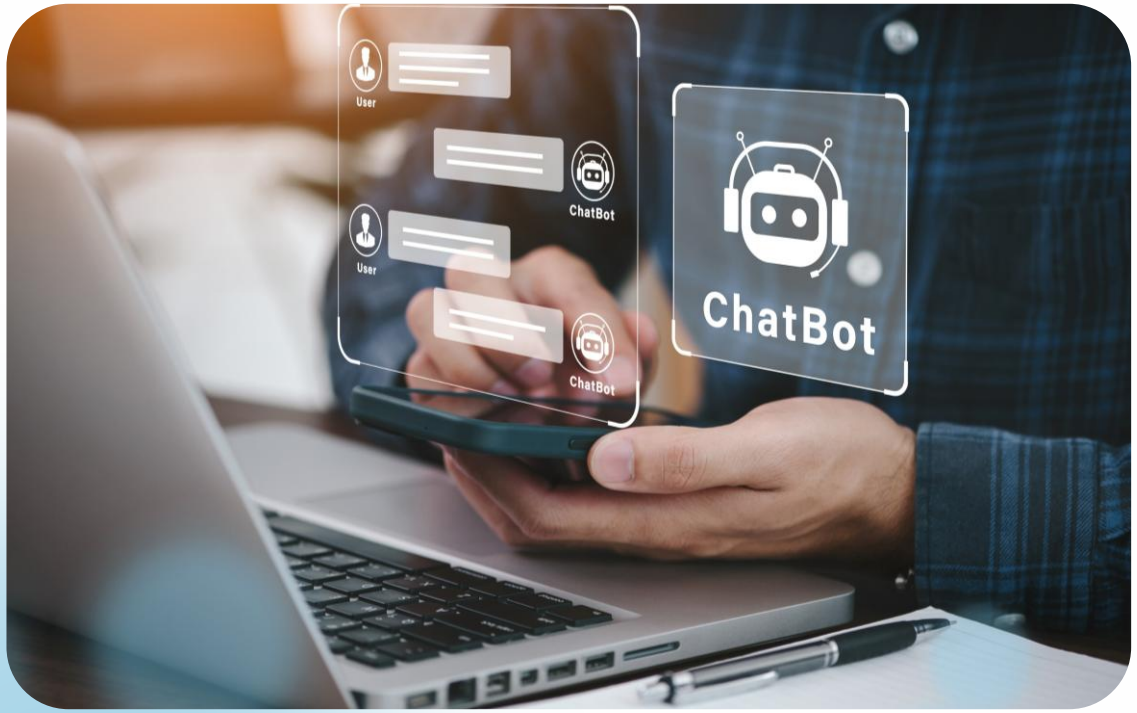
Provide clear, structured instructions to guide model behaviour and output format.

Response parsing

Extract usable data in formats like JSON, tables, or structured text from model outputs.

Prompt Formatting

What is Prompt Formatting



Strategic input design
Structure instructions, context and constraints to guide LLM behaviour

Clarity through structure
Set clear expectations and formats to reduce ambiguity and increase predictability

Human communication principles
Ask precise, well-structured questions to elicit actionable responses

Building Blocks of Effective Prompts

Context
Provide background, domain knowledge and situational awareness to frame the task.

Instruction
Give clear, specific directions including constraints and requirements.

Format specification
Define explicit guidelines for output structure, format and organisation.

Examples
Offer few-shot samples demonstrating the desired style, format and detail.

Prompt Example – Weak vs Strong

Poor prompt

"Answer the employee’s question about HR policies."

- Instructions lack clarity
- No output format specified
- Output length varies
- Hard to parse programmatically

Prompt Example – Weak vs Strong

Better prompt

"You are an internal HR Policy Assistant.
Use ONLY the policy documents provided in the CONTEXT section below.
For the given QUESTION, do the following:

1. Provide a short, clear answer in 2–4 sentences.
2. Cite the specific policy section(s) or clause IDs you used.
3. If the policy does not explicitly cover the question, say so clearly instead of guessing.
4. Do not invent policies or benefits that are not stated in the context.

CONTEXT:
{retrieved_policy_chunks}
QUESTION:
{employee_question}"

- Defined output format
- Fixed length guideline
- Organised structure
- Easily parseable results

Prompt Engineering Techniques

Zero-shot prompting

Give direct instructions without examples for simple, well-defined tasks

Few-shot prompting

Provide 2–5 input-output examples to guide format and style

Chain-of-thought prompting

Request step-by-step reasoning to enhance accuracy in complex tasks

Schema-guided prompting

Specify exact output formats using JSON schemas, CSV templates or Markdown structures for consistent parsing

Prompt Formatting Best Practices

Be specific
Define what you want, how you want it and any constraints to ensure consistency.

Set clear boundaries
Specify output length, format, and structure to simplify parsing and reduce complexity.

Use consistent language
Maintain consistent terms and phrasing to improve reliability and clarity

Test & iterate
Refine prompts continuously based on model outputs to optimise results

Response Parsing

What is Response Parsing

Extracting Structure from Chaos

Transforms LLM's free-form natural language output into structured, machine-readable data formats

Enables seamless integration with APIs, databases, analytics platforms and user interfaces

Supports the shift from experimental prototypes to production-ready applications

Response Parsing Approaches

Simple text cleaning
Use regex and string matching to extract information from unstructured responses

Structured output requests
Specify desired formats such as JSON, YAML or Markdown tables in the prompt

Parser libraries
Employ frameworks like LangChain's OutputParser, Guardrails AI or Pydantic for validation

Post-processing logic
Conduct validation, normalisation and error handling to ensure data integrity

JSON Output Parsing Example

Structured prompt: Extract company name and revenue from the text below. Respond in JSON format only with keys 'company' and 'revenue

Clean response:

```
{  "company": "OpenAI",  "revenue": "$1B+"}
```

Structured Output for Downstream Use

Ask the model to respond in strict JSON:

```json
{
  "short_answer": "<2-4 sentence answer>",
  "policy_citations": [
    {"document": "Leave_Policy.pdf", "section": "3.2"},
    {"document": "Contract_Benefits.pdf", "section": "1.4"}
  ],
  "needs_escalation": true,
  "escalation_reason": "Policy does not cover contract staff in probation period."
}
```

JSON Output Parsing Example

Integration Benefits

Parses easily into Python dictionaries

Inserts directly into SQL databases

Formats seamlessly for API responses

Formats seamlessly for API responses

Common Parsing Challenges

Format inconsistency
Models may ignore format rules or add unexpected text.

Incomplete structures
Missing fields, malformed JSON or inconsistent data disrupt parsing.

Validation failures
Outputs may not match schemas or contain invalid values.

Robust solutions
Use schema enforcement, retries, fallback parsing and thorough validation.

Production Workflow

Formatted prompt
Design structured input with clear format specifications and examples.

Structured response: Generate output following defined schemas (JSON, CSV, Markdown).

Parse & validate: Extract clean data using parsers with error handling and validation.

System integration: Integrate processed data into applications, databases, APIs or user interfaces.

Real-World Applications

Business Intelligence

Automated report summarisation with KPI extraction for executive dashboards and performance monitoring.

Conversational APIs

Structured chatbot responses integrated with business systems and customer service workflows.

Real-World Applications

Data processing

Automated labelling, classification and entity extraction for large data pipelines.

RAG systems

Controlled formatting of retrieved answers in Retrieval-Augmented Generation with consistent structure.

Master these techniques to build reliable, production-ready AI systems that integrate seamlessly with your infrastructure.

LangChain Tools & Tool Calling

Enhancing LLMs with External Capabilities

The Challenge: LLM Limitations

Large Language Models are powerful but face key constraints limiting real-world use.

Mathematical reliability: Struggle with precise calculations and complex arithmetic operations

Knowledge cutoffs: Use outdated training data lacking current events and real-time information

Structured outputs: Finds difficulty in producing consistent, API-compatible responses

Solution: Extend LLMs with external tools - functions, APIs and utilities that bridge these capability gaps.

What are LangChain Tools

Tools in LangChain are predefined or custom functions that an LLM calls to extend capabilities beyond text generation.

Mathematical operations
Precision calculators for arithmetic, algebra and complex computations

Information retrieval
Search engines and web scrapers for real-time, up-to-date data access

Data integration
Direct connections to databases, APIs, and enterprise systems

Custom logic
Business-specific functions and workflows tailored to your needs

Tool Calling Flow Architecture

User prompt
Submit question or request to the LLM

LLM Analysis
Model determines if external tools are needed

Tool execution
Send structured input to the selected tool

Result integration
Incorporate tool output into the final response

This workflow lets LLMs autonomously decide when and how to use external capabilities.

Example: Mathematical Precision

User prompt:
"What is (345 × 23) + 99?"

Pattern recognition
LLM detects a mathematical expression needing calculation

Tool invocation
Call calculator tool with structured input: "(345*23)+99“

Precise computation
Tool returns the accurate result: 8,034

Natural response
LLM formats the answer: "The result is 8,034"

Tools eliminate calculation errors and provide reliable mathematical results every time.

Example: Real-Time Information

User query:
"Who is the current CEO of Google?"

Query analysis
LLM recognises need for current information beyond training data

Search tool activation
Web search tool queries for "current Google CEO“

Data retrieval
Tool returns: "Sundar Pichai“

Contextual response:
"The current CEO of Google is Sundar Pichai"

Code Implementation Setup

Initialising LangChain with Multiple Tools for Enhanced LLM Capabilities

```python
# Import necessary modules
from langchain_openai import ChatOpenAI
from langchain.agents import initialize_agent, load_tools

# Initialize the LLM with deterministic behavior
llm = ChatOpenAI(temperature=0)

# Load pre-built tools for search and mathematics
tools = load_tools(["serpapi", "llm-math"], llm=llm)

# Create an agent that can reason about tool usage
agent = initialize_agent( tools=tools, llm=llm,  agent="zero-shot-react-description")

# Execute a complex query requiring multiple tools
result = agent.run(
    "What is the square root of 256 plus the CEO of Google's first name?")

# Display the result
print(result)
```

This setup lets the LLM autonomously select and use tools.

Observing Multi-Tool Execution

Demonstrating Sophisticated Reasoning in Processing Complex Queries

Recognises "square root of 256" requires calculator tool

Result: $\sqrt{256} = 16$

Mathematical analysis

Identifies need for current CEO information

Result: "Sundar" (first name)

Information lookup

Combines numerical and text results

Final: "16 + Sundar"

Intelligent synthesis

This shows the agent’s ability to coordinate multiple tools within one workflow.

Why Tool Calling Transforms AI

Action-oriented intelligence: Turns LLMs from text generators into problem solvers

Reduced hallucinations: Bases responses on verified external data

Real-world integration: Links AI systems to live data and operations

Domain expertise: Supports industry-specific, tailored workflows

Tool calling connects AI capabilities with real-world business applications.

Real-World Applications

Powerful, Domain-Specific AI Applications Enabled by Tool Calling

Enterprise integration

Direct queries to CRM, ERP and business intelligence systems for real-time insights and automated reports.

Educational technology

Intelligent tutoring systems offering step-by-step explanations with accurate calculations and verification.

Real-World Applications

Powerful, Domain-Specific AI Applications Enabled by Tool Calling

Research & analytics

Automated data retrieval, analysis and insights from live data sources and scientific databases.

Process automation

On-demand execution of scripts, workflows and business processes via natural language commands.

Demo

Introduction to Prompt Templates

Structuring prompts for reusability and building intelligent agents

Building on What We Know

Well-Formatted Prompts Improve LLM Responses, but Production Presents New Challenges

Reuse prompts across different tasks and contexts

Insert dynamic variables such as {question} and {context}

Ensure consistency while supporting flexibility

Prompt Templates provide the structured solution we need.

What is a Prompt Template

Use a structured prompt framework with placeholder variables filled dynamically at runtime

Replace variables such as {context} and {question} with actual values during execution

Ensures standardised instructions while adapting to varied inputs and use cases

Reusable pattern

Dynamic content

Consistency + Flexibility

Example prompt:
"Answer the following question based on the provided context:\n Context: {context}\n Question: {question}\n Answer:"

Why Prompt Templates Matters

Consistency: Standardise instructions and formatting across tasks to reduce variability and enhance reliability.

Reusability: Use one template to handle thousands of varied queries.

Control: Enforce specific response structures, tone and style while allowing flexibility for different contexts.

Integration: Pass dynamic variables from applications to LLMs for real-time, context-aware responses.

Practical Example: Q&A Template

```python
# Import the PromptTemplate class
from langchain.prompts import PromptTemplate
# Step 1: Define the template
template = """Answer the question based on the context below:

Context: {context}
Question: {question}
Answer:"""

# Step 2: Create the PromptTemplate
prompt = PromptTemplate(input_variables=["context", "question"], template=template)

# Step 3: Format the template with actual values
print(
    prompt.format(
        context="Paris is the capital of France.",
        question="What is the capital of France?" )
```

Output:
"Answer the question based on the context below: Context: Paris is the capital of France. Question: What is the capital of France? Answer: Paris"

Templates Power AI Agents

What are AI Agents?

Intelligent systems that combine LLMs with reasoning capabilities and external tools.

They can:

- Analyse problems step-by-step
- Choose appropriate tools
- Execute actions in sequence
- Synthesise final answers

Prompt templates guide their reasoning process, ensuring consistent decision-making patterns.

Templates Power AI Agents

Example

An agent uses templates to structure problem-solving, tool usage and result presentation

Building Your First Agent

Define available tools: Set up calculator and search functions your agent can use to solve problems.

Create reasoning template: Design a prompt guiding the model to think through problems systematically.

Initialise agent framework: Combine the LLM, tools and template into a cohesive, reasoning agent.

Execute and observe: Run queries and watch the agent analyse problems, utilise tools and produce answers.

Agent Implementation Code

```
from langchain_openai import ChatOpenAI
from langchain.agents import initialize_agent, load_tools
from langchain.prompts import PromptTemplate

llm = ChatOpenAI(temperature=0)
tools = load_tools(["serpapi", "llm-math"], llm=llm)

prompt = PromptTemplate(input_variables=["question"], template="You are a smart assistant
    . Use tools when needed.\nAnswer clearly and show your reasoning.\n\nQuestion:
    {question}\nAnswer:")

agent = initialize_agent(tools, llm, agent="zero-shot-react-description", verbose=True)

print(agent.run(prompt.format(question="What is the square root of 144 plus the current
    US President's first name?")))
```

Agent Reasoning Flow

Template processing

LLM reads the structured template: "You are a smart assistant. Use tools when needed...“

Tool selection & use

Agent identifies the need for calculator, computes √(144=12), then searches for current US President

Result synthesis

Combines mathematical result with search data to produce final structured answer

Reflect

Where Could I Use Agentic Tools in My Own Role/Workflow?

Identify tasks like summarising reports, drafting emails and cleaning data.

Spot decision-making steps for quick insights, such as market trends and financial projections.

Consider time-consuming research that agents could accelerate, like gathering industry benchmarks.

Explore workflows combining tools, for example, search and calculator for business analysis.

Ask, "*If I had a digital co-worker with access to search, memory, and reasoning, what tasks would I hand off?*"

Key Takeaways

- **Templates enable scale:** Prompt templates provide structure and reusability for consistent, production-ready AI systems.
- **Agents extend capabilities:** Combining LLMs, reasoning templates, and tools enables autonomous solving of complex, multi-step problems.
- **Foundation for innovation:** Simple agents highlight the synergy of prompts, reasoning and tools in AI workflows

Ready to build intelligent, scalable AI systems?

Q & A

Thank you